Highlights

Learnable Graph ODE Networks for Anomaly Detection in CAN-FD Vehicle Networks

Miao Xu, Lizeng Zhang, Peiyu Hou

- We combine dynamic time warping (DTW) with adaptive latent graphs to capture the physical dependencies (e.g., throttling - rotational speed - velocity chain) between the semantics of vehicular network messages and the evolving attack patterns, and address the limitations of previous work in modeling insufficient relationships between semantic time series data through iterative structural updates.
- We express spatiotemporal dynamics as a graphical ordinary differential equation (ODE) and combine it with a spatiotemporal graph ODE model (STGODE) predictor to learn the continuous changes in the relationships among semantic time series data, thereby achieving flexible predictions under different time conditions.

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Miao Xu[a], Lizeng Zhang[b] and Peiyu Hou[c]

[a]School of Civil Engineering and Transportation, Beihua University, Jilin, Jilin 132013, China.
[b]Haomo.AI Technology Co., Ltd, Beijing 101300, China.
[c]School of Civil Engineering and Transportation, Beihua University, Jilin, Jilin 132013, China.

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ABSTRACT

As vehicles become increasingly interconnected with networks, not only does communication between onboard systems grow more frequent, but real-time interactions between these systems and external infrastructure also significantly increase. Although these complex interactions promote the development of intelligent transportation, they also greatly increase the attack surface. In the face of such a background, identifying abnormal vehicle behavior in time becomes crucial to driving safety. The current anomaly detection methods cannot fully reflect the semantic information in the vehicle data. Without the knowledge of the logical relation of messages, models are very likely to produce a large amount of false positives, and it is difficult to explain the decision-making basis. To alleviate the above issues, we propose a learnable graph-based differential equation anomaly detection framework (LGOVAD) for CAN-FD vehicle networks. LGOVAD consists of three fundamental modules, which work cooperatively to interpret and model the semantic dynamics of vehicle communication data. The Semantic Extractor decodes interpretable physical signals based on the raw CAN messages. Then the Semantic Relationship Perception module discovers the hidden dependencies between vehicle parameters dynamically by employing a dual-branch graph learning strategy. Finally, we use a graph ODE network to capture the evolution process of these semantic patterns over time in a continuous manner. Experiments on real CAN-FD datasets demonstrate LGOVAD's ability to accurately identify semantic and temporal anomalies triggered by attack messages. Additionally, we validate the model's cross-domain generalization capabilities on SWaT and WADI datasets. Specifically, on CAN-FD data, LGOVAD achieves F1 scores 1.27-11.51% higher than specialized in-vehicle detectors (CANet, CANShield, PSEAD) and 3.48-16.42% higher than general multivariate temporal methods (GTA, FuGLAD). while maintaining robust advantages in normalized F1 scores on industrial benchmarks (SWaT, WADI).

1. Introduction

Modern intelligent connected vehicles (ICVs) combine cyber-physical systems, AI, and advanced networking to form complex computing platforms that enable autonomous driving and real-time vehicle-to-everything (V2X) interaction. The digital transformation of ICVs not only increases data exchanges between the in- and out-of-ICV worlds but also widens the attack surface from multiple heterogeneous interfaces. The Controller Area Network with Flexible Data-Rate (CAN-FD) is the backbone of real-time intercommunication between vehicle subsystems, responsible for transmitting physical parameters such as vehicle speed, brake pressure, and engine status. In contrast to conventional IT systems, the broadcast nature, lack of inherent authentication and reliability-over-security in CAN-FD create fundamental attacks that directly threaten physical safety.

Contemporary vehicle security approaches are subject to significant practical limitations. Cryptographic approaches, for instance, offer fundamental guarantees for confidentiality and integrity, yet they suffer from challenges in managing keys across heterogeneous ECUs, performance overhead in a real-time system, and inability to detect semantic attacks using authorized credentials [26]. These challenges drive the need for the development of advanced anomaly detection approaches capable of detecting malicious behaviors without architectural modifications and unacceptable latency. Recent advances in adjacent domains demonstrate alternative approaches: blockchain-based solutions have shown promise for ensuring data integrity in distributed robotic systems [1], while automated machine learning techniques have achieved remarkable accuracy in IoT intrusion detection [29]. However, these approaches require substantial adaptation to meet the unique constraints of vehicular networks, particularly the need for low-latency processing and semantic-aware anomaly detection.

Anomaly detection-based methods can detect and predict abnormal behaviors for responding to possible attacks in time [29]. In this sense, anomaly detection-based methods are widely used in anomaly detection in vehicular networks. The mainstream detection methods include detection methods based on network behaviors and deep learning-based detection methods. Network feature-based approaches employ in-vehicle network features for anomaly detection, which can be again subclassed into norm-based [17], ECU fingerprint-based [6], parameter monitoring-based [2, 13, 17] and information entropy-based [31], respectively. These approaches need manual feature extraction and mostly ignore the message contents, which are vulnerable to message tampering attacks and decrease the accuracy [26]. A key

Miao Xu(e-mail: xumiao@beihua.edu.cn) (M. Xu)
ORCID(s): 0009-0008-9102-7298 (M. Xu); 0009-0009-6046-8362 (L. Zhang); 0009-0008-8118-1988 (P. Hou)

limitation of approaches based on network-level anomaly detection is that they are all syntactic, i.e., focusing on the communication patterns without understanding the physical meaning behind the data. These methods work directly on the syntax of the network traffic: timing, frequency, or simple structural aspects; thus, they are all susceptible to semantic attacks that can manipulate the physical interpretation of the vehicle parameters while remaining compliant with the protocol. This is an important vulnerability of the security posture: the most dangerous attacks can be executed directly on the physical behavior and safety of the vehicle, through the manipulation of a semantically meaningful parameter, which is not obvious and is not an attack on the network traffic.

Deep learning-based anomaly detection techniques are gaining increasing attention in the field of vehicle network anomaly detection due to their powerful automatic feature extraction capabilities. Based on their detection scope, these methods can be broadly categorized into two types: data-domain-based approaches and semantic-based approaches. Tariq et al. [24] employed an RNN model using CAN message payload and ID as input features for in-vehicle intrusion detection. Qin et al.[19] applied LSTM-based time series prediction to CAN bus anomaly detection, achieving initial success in identifying payload anomalies through independent modeling of message IDs. Although data-domain-based methods can effectively detect malicious activities occurring within message data fields, their reliance on raw binary data results in insufficient detection granularity. In contrast, semantics-based methods achieve finer-grained detection capabilities. Hanselmann *et al.* [8] proposed CANet, an unsupervised autoencoder framework that handles high-dimensional CAN data for payload anomaly detection without labeled data. Jeong et al. [12] provide an interpretable attack diagnosis in CAN-FD through a Bi-LSTM network. Zhao et al. [33] pioneered attention-enhanced LSTM networks to detect physical semantic anomalies in CAN-FD messages, achieving granular analysis of time-series signal patterns at the application layer.

However, despite these advancements, current approaches suffer from several fundamental limitations:

- **Independent modeling of message IDs fails to capture inter-semantic dependencies and collaborative patterns.** Most semantic-based approaches decompose the CAN-FD time series by message ID and model each ID independently, often using separate recurrent or convolutional subnetworks [8, 12]. This design implicitly assumes that each message evolves in isolation, ignoring the relationships and interactions between semantics.

- **Lack of dynamic temporal structure modeling limits the detection of abnormal driving patterns**. Many prior models apply fixed window sizes or simple recurrent units to capture time dependencies, which fail to accommodate the highly dynamic, irregular, and long-range temporal dependencies in vehicular communication data [15, 8]. In diverse traffic scenarios, this flaw can lead to degraded anomaly detection performance.

- **Insufficient graph structure depth hinders long-range dependency modeling**. Existing graph neural network methods for time series analysis [4, 3] suffer from over-smoothing when stacking multiple layers, preventing them from capturing deep spatiotemporal dependencies essential for complex anomaly detection.

To address the issues discussed above, we develop Learnable Graph ODE Networks for Anomaly Detection in CAN-FD Vehicle Networks (LGOVAD), which leverages a combination of semantic feature engineering and high-level spatiotemporal modeling. The main contributions of LGOVAD are as follows:

- We propose a new semantic relationship perception module that learn dynamic graphs adaptively across semantics using dual branch graph learning, modeling explicit physical and latent interdependence among vehicle signals and their temporal evolution.

- We utilize the STGODE framework [5] as our spatiotemporal predictor, which formulates the dynamics as a graphical ordinary differential equation to learn continuous evolution of semantic relationships and thus to flexibly predict under varying temporal conditions, with a reduced over-smoothing problem in the conventional GNNs.

- We design an iterative graph fusion mechanism that dynamically balances prior knowledge from temporal alignment with data-driven relationship learning, enhancing robustness against incomplete parsing rules and adversarial manipulation in real vehicle environments.

- Extensive experimentation on real vehicle CAN-FD datasets demonstrates that LGOVAD consistently outperforms state-of-the-art baseline methods, achieving the highest F1 score (98.03% on SynCAN) while maintaining strong generalization capabilities across industrial control systems (94.62% on SWaT, 93.51% on WADI).

The following sections of this paper are organized as follows: Section 2 reviews the related work. Section 3 details the problem definition. Section 4 presents the motivation behind this research. Section 5 details the LGOVAD approach. Section 6 discusses performance evaluation through experiments. Finally, Section 7 summarizes the findings and explores future work.

2. Related Work

2.1. Graph neural networks-based Anomaly Detection in CAN-FD Vehicle Networks

GNN can capture graph-structured dependencies by propagation of the message across nodes, and thus models complex relationships between nodes. Thus, graph neural networks are often used to model interdependence between nodes with mutually dependent features. Recent studies have introduced graph-structured data to describe the complex relationships and interaction patterns within in-vehicle network traffic. Zhang et al. [32] built directed attributed graphs from CAN message flows within a given message period and exploited GNNs on the messages for anomaly detection classification. Islam et al. [10] leveraged the graph data structure on CAN message time windows and detected abnormalities with graph property distributions. xiao et al. [28] transformed the temporal relationships of CAN messages into graph structure data and employed a graph attention mechanism to detect anomalies. yang et al. [30] proposed CAN-ASTGCN, an attention-based spatio-temporal GCN framework for CAN anomaly detection that integrates temporal modeling. Wang et al. [25] designed a dynamic graph-based semi-supervised anomaly detector for CAN networks, leveraging temporal graph attention to track structural evolution without relying on ECU identifiers. Sun et al. [23] constructed an elevator anomaly detection model using CAN bus data without additional sensors, applying GNNs and Gini index-based anomaly scoring to recognize unseen failure types. Luo et al. [16] introduced CA-GCN, a causality-aware GCN framework that models semantic interactions in vehicular networks by combining semantic extraction, transfer-entropy-based causal reasoning, conditional mutual information filtering, and temporal graph convolution.

However, these models mostly use the message sequence relations as graph edges and utilize the raw binary data of the message as node features. Therefore, they are all still in the framework of data domain anomaly detection without modelling the rich semantic relations among message content.

2.2. Multivariate Time Series Anomaly Detection Method

Since the semantic features of in-vehicle networks are essentially multivariate time series data (MTS), the semantical-based methods can be regarded as the multivariate time series anomaly detection (MTSAD) techniques[16]. Therefore, in this subsection, we discuss the current MTSAD method. LSTM-DAE [11] employs the first-stage to select outlier candidates based on signals from each operation cycle, and the second-stage to analyze the sensor data with temporal coherence, to detect abnormal states. LSTM-VAE [18] integrates LSTM and VAE architectures to capture the temporal dynamics and perform probabilistic reconstruction, which can be used to detect abnormal states through reconstruction error. However, these methods focus on temporal features, which usually neglect to model the interrelationship in multidimensional time series, especially without considering the possible functional interactions. To mitigate this, graph neural networks (GNNs) have been employed to model relationships within MTS, leveraging their proficiency in capturing graph structural relationships. GDN [4] leverages graph neural networks to model dependencies between time series variables for improved anomaly detection. FuSAGNet [7] introduces fusion sparse self-encoder graph modeling univariate timing relationships, and combines a dual-channel mechanism of prediction and reconstruction to detect anomalies. GTA [3] designs a full-parameter graph learning strategy that integrates graph convolution and extended convolution to form a hierarchical context coding block based on the Transformer to achieve timing deviation diagnosis. FuGLAD [9] utilizes a priori graph-based fusion method to fuse multi-graph structures to describe complex topological relationships of sensors.

However, these methods fail to fully utilize the known information, i.e., the information contained in the time series data itself. In addition, existing graph neural network methods are insufficient in capturing long-distance dependencies between MTS.

3. Problem Definition

Given a collection of raw in-vehicle network messages $\mathrm{M}_{\mathrm{train}} = \left[\mathrm{m}_1, \mathrm{m}_2, \ldots, \mathrm{m}_{\mathrm{n}_{\mathrm{train}}}\right]^{\mathrm{T}}$ and a parsing rule base $\mathcal{R}$, the anomaly detection procedure initiates by generating semantic time series $X \in \mathbb{R}^{K \times N}$, where K represents the feature dimensionality and N signifies the number of timestamps. Subsequently, the model identifies anomalies within the resulting multivariate time series X. This is achieved through the following generalized process:

$$X = \Phi(\mathcal{M}, \mathcal{R}) \tag{1}$$

$$\hat{x}_t = f(X_t) \tag{2}$$

$$s_t = \varphi(x_t, \hat{x}_t) \tag{3}$$

$$\hat{y}_t = \begin{cases} 1, & \text{if } s_t > T \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

Where $\Phi(\cdot)$ denotes the semantic extraction and preprocessing function that converts raw CAN messages $\mathcal{M}$ into a normalized multivariate time series X, $X_t \in \mathbb{R}^{K \times \omega}$ is the historical subsequence (window of size ω) used for predicting the current value x_t, $f(\cdot)$ is the prediction model that estimates the expected value $\hat{x}_t$ at time t, $\varphi(\cdot)$ is the anomaly scoring function (e.g., absolute error, Mahalanobis distance, etc.), T is a predefined or learned threshold, $\hat{y}_t$ is the binary anomaly label.

4. Motivation

The rapid advancement of ICVs has significantly expanded the attack surface of in-vehicle networks, making robust anomaly detection crucial for ensuring vehicle safety. The CAN-FD bus, serving as the central nervous system

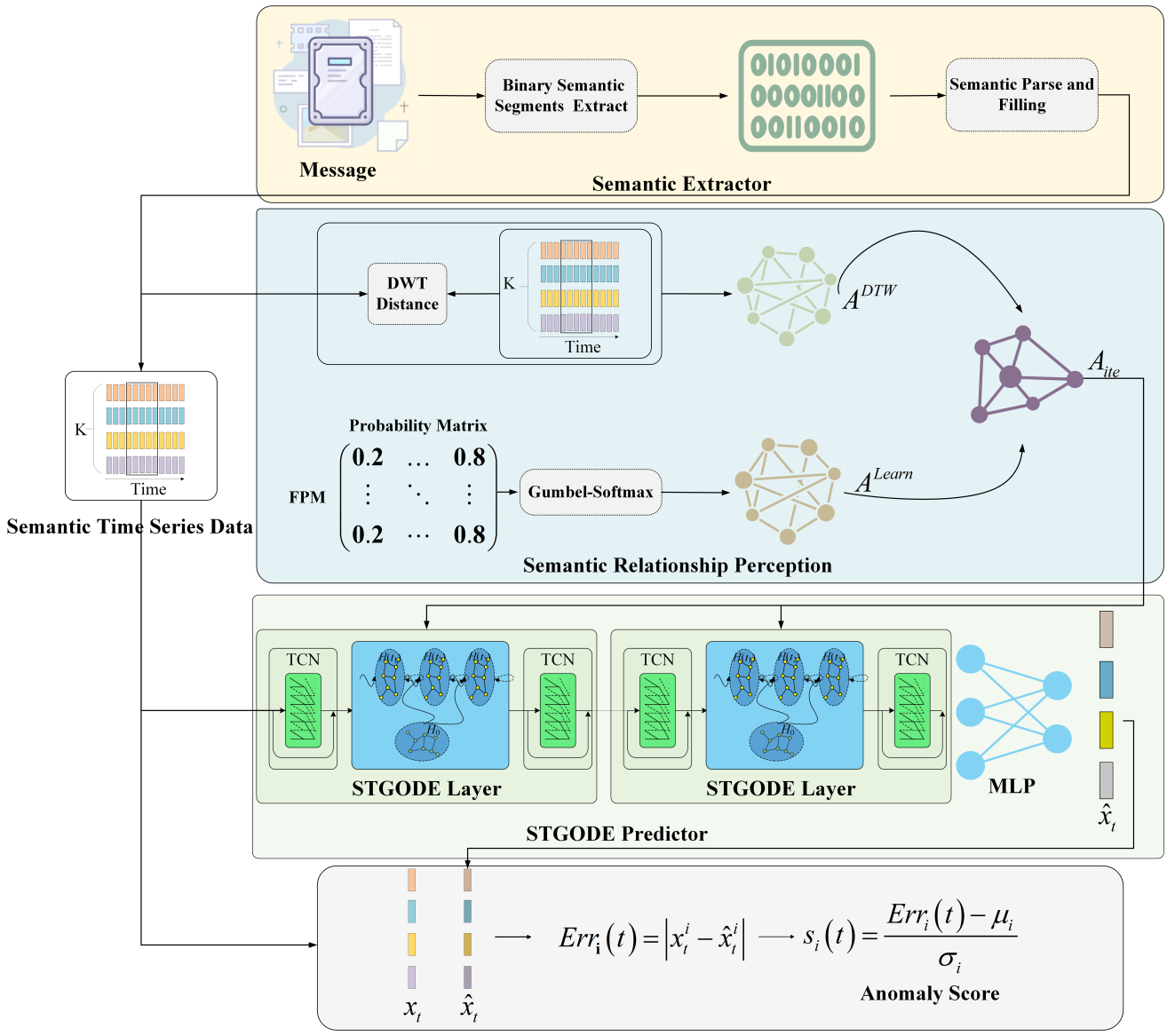

Figure 1: The framework of LGOVAD

for real-time communication among critical subsystems, has become a primary target for malicious attacks through various external interfaces.

Current anomaly detection methods have two fundamental limitations to work effectively in real-world vehicle environments. First, data domain-based methods working with the raw binary payloads do not have any understanding of the semantics, which leads to coarseness of detection granularity and limited detection accuracy on sophisticated attacks. Second, while semantic-based methods generally learn each signal separately, the complex physical dependencies between vehicle parameters (e.g., the cause-and-effect chain from throttle position to engine RPM to vehicle speed) are not well modeled. Third, current temporal modeling methods with fixed window sizes or simple recurrent units cannot handle highly dynamic and disordered temporal dependencies in vehicular communication data. The rigidity makes the model perform poorly under a wide variety of driving conditions and is less capable of detecting changing pattern attacks.

To fill these crucial gaps, we are inspired to design LGOVAD, which unifies the semantic-aware feature engineering and spatio-temporal continuous modeling. Specifically, we attempt to explicitly learn the dynamic relationships between the semantic signals and model the changes over time to detect sophisticated anomalies, which often emerge in abnormal semantic temporal patterns.

5. Proposed Methodology

5.1. Overview

We introduce LGOVAD to address the complex dependencies in the modeling of the vehicular network time series and detect anomalies. Fig. 1 presents the overall framework

of LGOVAD, which can be summarized as a combination of semantic-aware feature engineering and continuous spatio-temporal modeling. It consists of three key components:

- **Semantic Extractor:** It converts raw in-vehicle CAN messages $\mathcal{M}$ into normalized semantic time series $X \in \mathbb{R}^{K \times N}$.
- **Semantic Relationship Perception:** It learns temporal and semantic feature dependencies via dual-branch graph learning.
- **STGODE Predictor and Anomaly Score:** It models continuous spatio-temporal dynamics and quantifies anomalies.

The overall process of LGOVAD is summarised in Algorithm 1. Raw CAN messages are parsed and normalized into time series by the semantic extractor. The graph based on DTW distance and the learnable latent graph is utilized by LGOVAD to update $A^{(t)}$ in every step. The value $\hat{x}_t$ will be predicted in the next step by the STGODE module based on X_t and $A^{(t)}$. The anomaly score of the standardisation is calculated based on the residual between the predicted value and the ground truth. A grid search-based algorithm is employed to select the detection threshold that works the best.

5.2. Semantic Extractor

The semantic extractor converts raw in-vehicle network messages $\mathrm{M}_{\text{train}} = \left[\mathrm{m}_1, \mathrm{m}_2, \ldots, \mathrm{m}_{\mathrm{n}_{\text{train}}}\right]^{\mathrm{T}}$ to semantic time series data $\mathrm{s} = \left(\mathrm{s}_1, \mathrm{s}_2, \ldots, \mathrm{s}_{\mathrm{n}_{\mathrm{s}}}\right)^{\mathrm{T}}$. In our analysis, a resampling strategy is used to accommodate heterogeneous signal update rates while reducing information loss. We start by analyzing a general scenario and use the following formal analysis.

- **Signal Characteristics Analysis**: A vehicle network signal might update at different time scales. The signal with critical controls (e.g., throttle position, brake pressure) is often at 10-100Hz, or even 1-10Hz for diagnostic and status signals. We choose our resampling interval t = 10ms to meet the Nyquist criterion for the highest-frequency signals, but not introduce too much overhead.
- **Aliasing Prevention**: For signals with update frequencies f_s, the resampling interval Δt ensures that the effective sampling rate $f_{\text{resample}} = 1/\Delta t = 100\text{Hz}$ satisfies $f_{\text{resample}} > 2 \cdot \max(f_s)$ for all vehicular signals, preventing aliasing according to the Nyquist-Shannon sampling theorem.
- **Information Loss Minimization**: The forward-filling imputation strategy preserves the most recent valid measurement until a new update arrives. This approach is mathematically equivalent to a zero-order hold (ZOH) reconstruction, which minimizes high-frequency artifacts while maintaining signal integrity for anomaly detection purposes.
- **Multi-rate Signal Handling**: The unified timeline $\mathcal{T}$ accommodates signals with different native update rates through the following formulation:

$$S_j[t_k] = \begin{cases} v_{\text{current}} & \text{if } \exists \text{ update at } t_k \\ S_j[t_{k-1}] & \text{otherwise} \end{cases} \tag{5}$$

  where $S_j[t_k]$ represents the value of signal j at resampled timestamp t_k.
- **Temporal Consistency**: The inferred timing features (inter-message timing, message frequency) are complements for the values, which may help to identify if the signal evolution is normal or abnormal even when the signal values look reasonable.

As shown in Algorithm 2, the semantic extractor extracts semantics from vehicle network messages through the following steps:

Initially, for each message frame m within the dataset $\mathcal{M}$, the semantic extractor begins by identifying the unique identifier of the message ID_m. Subsequently, it retrieves the associated parsing rules $\mathcal{R}_m$ from the rule repository, using the identifier. These rules are then applied to extract the binary data segments that encode the semantic information within the message payload. While LGOVAD relies on parsing rules $\mathcal{R}$ for semantic extraction, the framework demonstrates inherent robustness to incomplete or erroneous rules through several design mechanisms. First, the semantic extractor incorporates temporal features (inter-message timing and frequency statistics) that remain valid even when payload parsing rules are incomplete, providing complementary signals. Second, the dual-branch graph learning compensates for potential rule errors by adaptively learning latent relationships between signals that may not be captured by predefined rules. For instance, if throttle position rules are erroneous, the learnable graph component can still capture its correlation with RPM through temporal patterns. Third, the iterative graph update mechanism (Eq. 14) allows the model to progressively refine relationships despite initial rule inaccuracies. The raw data segments undergo parsing and transformation into interpretable signals. Simultaneously, the ID and other attributes of the message frame, including the inter-message time, are encoded as supplementary semantic features to augment the information representation capacity, as demonstrated in lines 2-7 of the Algorithm 2. After semantic feature extraction, the features are categorized by type. For any semantic features absent from the current message frame, the parsing values from the most recent frame containing the feature are employed to impute the missing data, thereby generating the semantic time series data s, as demonstrated in lines 8-11 of the Algorithm 2. To ensure comparability, we normalized the semantic feature dimensions. The normalization procedure is detailed below:

Algorithm 1: LGOVAD for Vehicle Network Anomaly Detection

Input : $\mathcal{M}$: Raw CAN message dataset
$\mathcal{R}$: Parsing rule base
ω: Sliding window size
λ: Graph mixing factor
τ_{init}: Initial threshold (0.5)
ϵ_{DTW}: DTW sparsity threshold (0.1)
Output: $\mathcal{A}$: Anomaly timestamps

$X \leftarrow [\,]$; // Initialize normalized time series tensor
$\mathcal{S} \leftarrow$ Semantic Extraction$(\mathcal{M}, \mathcal{R})$; // See Algorithm 2
for *each semantic signal* $s_k \in \mathcal{S}$ **do**
  $x_k \leftarrow$ Dynamic Normalization(s_k) ; // Eq. 1-2 with $\epsilon = 0.01, \mu = 1$
  $X \leftarrow X \oplus x_k$; // Append to feature tensor
$\mathcal{W} \leftarrow$ Sliding Window Partition(X, ω) ; // $\mathcal{W} \in \mathbb{R}^{T \times K \times \omega}$
$T \leftarrow |\mathcal{W}|$; // Total timesteps
$A^{\text{DTW}} \leftarrow$ Compute DTW Graph$(X, \epsilon_{\text{DTW}})$; // Eq. 4-5
$\pi \leftarrow \text{Uniform}(0,1)^{K \times K}$; // Learnable probability matrix
$A^{\text{learn}} \leftarrow$ Gumbel-Softmax Sampling(π) ; // Eq. 6-7 with $\tau = 0.5$
$A^{(0)} \leftarrow A^{\text{DTW}}$; // Initialize graph structure
$\mathcal{S} \leftarrow [\,]$; // Anomaly score sequence
for $t \leftarrow 1$ **to** T **do**
  $A^{(t)} \leftarrow \lambda A^{(t-1)} + (1-\lambda) A^{\text{learn}}$; // Eq. 9
  $X_t \leftarrow \mathcal{W}[t]$; // Current window $[x_{t-\omega}, ..., x_{t-1}]$
  // STGODE Prediction (Eq. 10-14)
  $\hat{\mathcal{H}}_0 \leftarrow \text{TCN}(X_t)$; // Temporal feature extraction
  $\hat{\mathcal{H}}(\varepsilon) \leftarrow \text{ODESolver}\left(\frac{d\mathcal{H}}{d\varepsilon}, \hat{\mathcal{H}}_0, \varepsilon\right) \hat{x}_t \leftarrow \text{MLP}(\hat{\mathcal{H}}(\varepsilon))$; // Next-step prediction
  // Anomaly Scoring (Eq. 15-18)
  $s(t) \leftarrow$ Anomaly Scoring$(\hat{x}_t, x_t)$; // Max standardized error
  $\mathcal{S} \leftarrow \mathcal{S} \oplus s(t)$
$\tau_{\text{opt}} \leftarrow$ Grid Search Threshold$(\mathcal{S})$; // Maximize F1 score
$\mathcal{A} \leftarrow \{t \mid s(t) > \tau_{\text{opt}}, \forall t \in [1, T]\}$
return $\mathcal{A}$

$$x_i = \begin{cases} 2s_i - s_{i,\max} - s_{i,\min} & \text{if } s_{i,\max} - s_{i,\min} \geq \epsilon \\ 2s_i - s_{i,\max} - s_{i,\min} - \mu & \text{if } s_{i,\max} - s_{i,\min} \leq \epsilon \end{cases} \tag{6}$$

$$x_i = \max\left(\min(x_i, x_{\max}), x_{\min}\right) \tag{7}$$

where $s_{i,\max}$ and $s_{i,\min}$ represent the maximum and minimum values of the semantic feature s_i in the training set, respectively; When the feature value range $s_{i,\max} - s_{i,\min} \leq \epsilon$, perform upper bound adjustment $s_{i,\max} \leftarrow s_{i,\max} + \mu$; The parameters are set as $\epsilon = 0.01$, $\mu = 1$, $x_{\min} = -2$, and $x_{\max} = 2$.

Finally, for standardized semantic time series data $X = (\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)^T \in \mathbb{R}^{K \times N}$, where K denotes the feature dimension and N the number of timestamps, we partition the data using sliding windows of length ω. The historical window at timestamp t is defined as the subsequence $X_t = (\boldsymbol{x}_{t-\omega}, \boldsymbol{x}_{t-\omega+1}, \ldots, \boldsymbol{x}_{t-1})^T \in \mathbb{R}^{K \times \omega}$. Fig. 2 shows the overall workflow of the semantic extractor.

5.3. Semantic Relationship Perception

In the vehicle network, the semantic information (vehicle speed, engine speed, throttle value, etc.) has complex causal relationships. For example, the increase in throttle value will cause an increase in engine speed and then affect the vehicle speed. Modeling and analyzing these relationships enables the model to capture semantic timing relationships and abnormal change behaviors, thus improving the efficiency of anomaly detection in the vehicle network. To model the semantic time series relationships, we propose a dual-branch graph structure learning method (DGL) by mapping the semantic features to the nodes while mapping the relationship of semantic features into edges, which learns the graph structure with the feature relationships. As illustrated in the blue part of Fig. 1, DGL simultaneously employs the DWT distance between the raw STS and the direct graph structure learning method to learn the semantic relationship in a recursive way. Specifically, this architecture mainly consists of the following three steps.

Algorithm 2: Enhanced Semantic Time Series Extraction with Multi-rate Signal Handling

Input : $\mathcal{M} = \{(t_i, id_i, payload_i)\}_{i=1}^{n}$: Timestamped CAN messages
$\mathcal{R}$: Parsing rule base
Δt: Resampling interval (e.g., 10ms)
$\mathcal{ID}_{\text{target}}$: Critical message IDs (e.g., 0x304)
Output: S: Normalized semantic time series

```
S_raw ← ∅ ;                                          // Temporary storage for parsed messages
last_time ← { } ;                                    // Dictionary: last occurrence time per ID
signal_registry ← ∅ ;                                // Set of all semantic signal names
for each message (t, id, payload) ∈ M do
    R_m ← load deserialization rules(R, id) ;
    {p_1, p_2, ..., p_k} ← decode payload segments(payload, R_m) ;
    (s_1, s_2, ..., s_k) ← convert to physical values({p_1, ..., p_k}, R_m) ;
    if id ∈ last_time then
        δt ← t − last_time[id] ;
        append ("inter_time_" + id, δt) to signals
    last_time[id] ← t ;                              // Update last seen time
    if id ∈ ID_target then
        append ("msg_count_" + id, 1) to signals
    signal_registry ← signal_registry ∪ {signal names} ;
    append (t, {(name, value)}) to S_raw ;
t_min ← min{t_i}, t_max ← max{t_i} ;
T ← [t_min, t_min + Δt, ..., t_max] ;                // Regular time grid
S_aligned ← array of NaNs with dimensions |T| × |signal_registry|
for each signal s_j ∈ signal_registry do
    t_prev ← NaN, v_prev ← NaN ;
    for each timestamp t_k ∈ T do
        if ∃(t_k, values) ∈ S_raw where s_j ∈ values then
            v_prev ← value of s_j at t_k ;
        S_aligned[k, j] ← v_prev ;                   // Forward filling (zero-order hold)
S ← add timing features(S_aligned) ;                 // Add: msg frequency, time since last update
return S
```

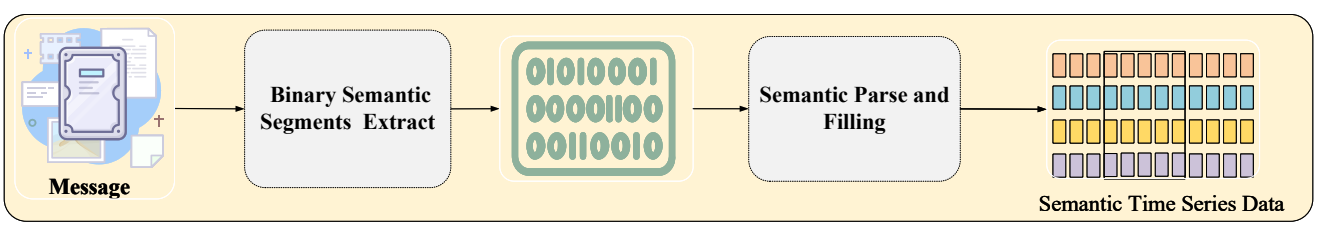

Figure 2: workflow of the semantic extractor

5.3.1. Direct Semantic Relation Learning

Given two MTS $T = (t_1, t_2, ..., t_n)$ and $U = (u_1, u_2, ..., u_m)$, a sequence distance matrix $D \in \mathbb{R}^{n \times m}$ is defined, where each element $D_{i,j} = |t_i - u_j|$. The DTW prior graph is formulated as follows:

$$D_c(i,j) = D_{i,j} + \min(D_c(i-1,j), D_c(i,j-1), D_c(i-1,j-1)) \tag{8}$$

$$DTW(T,U) = D_c(n,m) \tag{9}$$

$$S_{i,j}^{DTW} = DTW(x^i, x^j) \tag{10}$$

$$A_{i,j}^{DTW} = \begin{cases} 1, & S_{i,j}^{DTW} < \varepsilon \\ 0, & otherwise \end{cases} \tag{11}$$

where D_c is the cost matrix, $S^{DTW} \in R^{N \times N}$ is the similarity matrix obtained for all indicators based on DTW distance. A^{DTW} is the DTW prior graph adjacency matrix, ε

is a parameter to determine the sparsity of adjacency matrix A^{DTW}.

5.3.2. Latent Semantic Relationship Learning

In addition to the information contained in the STS itself, there may be some potential unknowns between features. To fully model the relationship between features, we further adopt a Full parameter learning method (FPM), treating the relationship between features as an unknown graph structure, and each element in the graph structure is a learnable parameter. However, the adjacency matrix is typically composed of 0 and 1, making gradient calculations challenging. To address this issue, Gumbel-Softmax is employed to convert discrete parameters of the adjacency matrix into continuous ones for backpropagation. The process of Gumbel-Softmax is defined as follows:

$$g = -\log(-\log(u)), u \backsim \text{Uniform}\,(0,1) \tag{12}$$

$$z_1^{i,j} = \frac{\exp\left(\left(\log \pi_1^{i,j} + g^{i,j}\right)/\tau\right)}{\sum\limits_{v\in\{0,1\}} \exp\left(\left(\log \pi_v^{i,j} + g^{i,j}\right)/\tau\right)} \tag{13}$$

where u represents the sample extracted from a Uniform(0,1) distribution, $g^{i,j}$ follows a Gumbel distribution, and $\pi_1^{i,j}$ denotes the values of the i-th row and j-th column of the probability matrix $\pi_1 \in R^{K\times K}$ which signifies the probability that node i is connected to node j in the graph. Here, τ serves as a temperature parameter. As τ approaches 0, $z_1^{i,j}$ tends towards 0 or 1, and the Gumbel-Softmax distribution converges to the class distribution. Ultimately, the i-th row and j-th column in the adjacency matrix $A_{i,j}^{Learn}$ are assigned the value of $z_1^{i,j}$, causing $A_{i,j}^{Learn}$ to be set to 1 with the probability of $\pi_1^{i,j}$.

5.3.3. Iterative Learning

To fully utilize the original STS information while considering the potential relationship between features, we first use the matrix A^{DTW} to initialize the graph structure and then update the structure based on the learned matrix A^{learn}. In this way, both intrinsic STS characteristics and the latent feature relationship are taken into consideration when updating the graph structure. The method formula is as follows:

$$A^{(t)} = \lambda \cdot A^{(t-1)} + (1-\lambda)\cdot A^{\text{learn}} \tag{14}$$

where $A^{(t-1)}$ denotes the graph structure at iteration $t-1$, and $A^{(0)} = A^{\text{DTW}}$ represents the initial graph structure constructed using DTW distance. This iterative update mechanism:

Through iterative refinement, this formulation progressively integrates both the temporal alignment information from A^{DTW} and the latent feature correlations captured in A^{learn}.

5.4. STGODE Predictor

To fully model dependencies among semantic features, we employ STGODE as the predictor. Compared to discrete GCNs, STGODE mitigates over-smoothing through its continuous-time dynamic mechanism. When using discrete GCNs, node representations rapidly converge to shared states under repeated aggregation operations, leading to over-smoothing. In contrast, STGODE employs a differential equation-based approach that models graph convolutions as continuous processes, enabling controllable information propagation through differential equations. This continuous integration prevents the rapid convergence by enabling smoother and more gradual updates to node representations, thus maintaining feature distinctiveness even at deeper equivalent depths (e.g., 8-10 layers). The ODESolver adaptively controls the integration step, ensuring that node features do not oversmooth while capturing long-range dependencies. The predictor accepts two inputs:

$$\mathcal{P}(A, X_t) \rightarrow \hat{x}_t \tag{15}$$

where $A = A^{(T)}$ denotes the final graph structure after T iterations of Eq. (14), integrating both A^{DTW} and A^{learn} information and $X_t \in \mathbb{R}^{N\times F}$ represents the multivariate time series subsequence at time t, with N nodes and F features.

The subsequence X_t undergoes processing through stacked STGODE layers to capture spatio-temporal dependencies. The resulting representation is then transformed via a multilayer perceptron (MLP) to produce the final single-step prediction output $\hat{x}_t$.

Traditional graph convolutional networks (GCNs) is a discrete format, which only aggregates n-order neighbors with n-layers. They lead to over-smoothing after multiple layer stacking and can not capture long-distance dependencies between sensors. To overcome these limitations, the STGODE Predictor uses a continuous GCN formulation based on ODEs [5].

The continuous GCN can be represented by ODE [27] as follows:

$$\mathcal{H}(\varepsilon) = \int_0^{\varepsilon+1} \mathcal{H}_0 \times_1 \widehat{A}^{\tau} \times_2 U^{\tau} \times_3 W^{\tau} d\tau \tag{16}$$

$$\widehat{A} = \frac{\alpha}{2}(I + D^{-\frac{1}{2}} A D^{-\frac{1}{2}}) \tag{17}$$

where $\mathcal{H}_0$ denotes the initial input, $\times_i$ denotes the tensor matrix multiplication on mode i, W and U are learnable parameter matrices, and s is a continuous variable of layer n in discrete GCN. D is the degree matrix of A, $\widehat{A}$ is the regularized adjacency matrix, and $\alpha \in (0,1)$ is a hyperparameter. The ODE formulation allows the model to capture long-distance dependencies and avoid over-smoothing by integrating information continuously across layers. The ODE can get an analytical solution by the ODESolver.

A STGODE block employs a "sandwich" structure, sequentially combining two TCN layers with an ODESolver. It can extract spatio-temporal features and capture long-distance dependencies.

$$H_{tcn}^{l} = \begin{cases} X_t & , l = 0 \\ \sigma(W^l *_{d^l} H_{tcn}^{l-1}) & , l = 1, 2, \cdots, L \end{cases} \tag{18}$$

$$\mathcal{H}(\varepsilon) = \text{ODESolver}\left(\frac{d\mathcal{H}(\varepsilon)}{d\varepsilon}, \mathcal{H}_0, \varepsilon\right) \tag{19}$$

where $\frac{d\mathcal{H}(\varepsilon)}{d\varepsilon} = \mathcal{H}(\varepsilon)\times_1(\widehat{A} - I) + \mathcal{H}(\varepsilon)\times_2(U - I) + \mathcal{H}(\varepsilon)\times_3(W - I) + \mathcal{H}_0$, X_t is the input of TCN, H_{tcn}^{l} is the output of the l-th layer of TCN, W^l denotes the l-th convolution kernel, $d^l = 2^{l-1}$ denotes exponential dilation rate.

5.5. Loss Function

The mean squared error (MSE) is employed as the prediction loss $\mathcal{L}_P$:

$$\mathcal{L}_P = \frac{1}{K}\sum_{i=1}^{K}\left(\hat{x}_t^{(i)} - x_t^{(i)}\right)^2 \tag{20}$$

where $\hat{x}_t^{(i)}$ and $x_t^{(i)}$ denote the predicted and ground-truth values of the i-th temporal variable at timestep t, respectively.

To prevent overfitting, an L_2 regularization term is incorporated, yielding the composite loss function:

$$\mathcal{L} = \mathcal{L}_P + \lambda|\mathbf{W}|_F^2 \tag{21}$$

where $\lambda \in \mathbb{R}^+$ governs regularization intensity, and $\mathbf{W}$ represents the trainable parameter matrix.

5.6. Anomaly Scores and Threshold Selection

Anomaly detection targets the identification of abnormal behavior of systems by comparing the predictions to ground truth. Our detection pipeline comprises three steps: calculating sensor-level anomaly scores, temporal aggregation, and threshold-based classification. We mark a data point at timestamp t anomaly if its overall anomaly score is above the threshold.

The prediction error for sensor i at time t is computed as:

$$\text{Err}_i(t) = \left|x_t^{(i)} - \hat{x}_t^{(i)}\right| \tag{22}$$

To enable cross-sensor comparison, prediction errors are standardized using sensor-specific statistics:

$$s_i(t) = \frac{\text{Err}_i(t) - \mu_i}{\sigma_i} \tag{23}$$

where μ_i and σ_i denote the mean and standard deviation of $\text{Err}_i(t)$ over the validation period.

The timestamp-level anomaly score aggregates sensor-wise deviations via maximum pooling:

$$s(t) = \max_i, s_i(t) \tag{24}$$

Table 1
Detailed information of datasets

Dataset	SWaT	WADI	SynCAN
Dimension	51	127	33
# of training	495000	762970	67200
# of testing	450000	172800	20893
Anomalies	11.97%	5.99%	19.59%

Threshold optimization is performed using a grid search across the domain $\mathcal{T} = [s_{\min}, s_{\max}]$, where $s_{\min}$ and $s_{\max}$ denote the minimum and maximum observed values of $s(t)$. The search evaluates threshold values $\tau \in \mathcal{T}$ at intervals of 0.01, selecting the optimal threshold τ^* that maximizes the F1-score:

$$\tau^* = \underset{\tau\in\mathcal{T}}{\text{argmax}}\left(F_1(\tau)\right)$$

We do this only on a held-out validation set consisting of only normal samples from the training distribution, i.e., no anomalies of the test data are seen, and the comparison remains unbiased. A point extension step of the detected anomalous regions is applied after thresholding to include sub-threshold points in the event boundaries to create contiguous regions.

6. Experiments Evaluation

This section provides a comprehensive analysis of our experiment.

6.1. Datasets

In the experimental section, we utilized a real-world vehicle SynCAN dataset along with two publicly available datasets, SWaT and WADI. The SynCAN dataset is used to evaluate LGOVAD's performance in detecting semantic anomalies in vehicle networks. Besides, the SWaT and WADI datasets were used to validate the general applicability of LGOVAD in the MTSAD task. The detailed statistical information for each dataset is summarized in Table 1.

- **Synthetic CAN Bus Data (SynCAN):**[1] SynCAN is a well-known public signal-level CAN attack dataset released by EATS, constructed based on real CAN trace characteristics. It contains a wealth of normal driving data and hundreds of advanced attack events, covering 20 signals and 10 CAN IDs. The dataset includes three main types of attacks: manufacturing attacks (i.e., flooding attacks, which by sending high-priority messages at high frequency within a short timeframe, legitimate ECU communications are disrupted.), suspension attacks (i.e., suppression attacks, which prevent the target ECU from sending messages on the bus by shutting it down), and spoofing attacks (which include platform attacks, continuous attacks,

[1] https://github.com/etas/SynCAN

and replay attacks, all involving shutting down the target ECU and sending malicious messages under its guise). Among these, platform attacks cause the target signal to broadcast a constant value during the attack, potentially leading to false transitions or freezing of actual values, with the impact depending on the transition amplitude and duration; continuous attacks and replay attacks, on the other hand, inject continuously changing or pre-recorded target signal values, with the signal initially approaching actual values but gradually deviating significantly and detectably over time. For the SynCAN dataset, we apply the same temporal alignment and imputation strategy described in Algorithm 2. Raw CAN messages are first resampled to a uniform 10ms interval to handle irregular message frequencies typical in vehicular networks. Missing signals are handled through forward-filling, where the last observed value is propagated until a new measurement arrives. This approach effectively handles scenarios where certain ECUs temporarily stop transmitting, such as during engine start-up sequences. For example, throttle position signals (message ID 0x100) may experience intermittent drops during rapid bus load changes; our forward-filling mechanism maintains signal continuity by preserving the last valid value until the next update.

- **Safe Water Treatment (SWaT) Dataset:**[2] SWaT dataset originates from a water treatment test bench supervised by the Singapore Public Utilities Bureau. Over 11 days of continuous 24-hour operation, network traffic and data from 51 sensors and actuators were recorded.
- **Water Distribution (WADI) Dataset:**[3] WADI is an extension of the SWaT platform, representing a more comprehensive water treatment, storage, and distribution system with 127 sensors and actuators. Data were collected over 16 days, with 14 days dedicated to normal operations and 2 days simulating attack scenarios.

To process raw data, the SWaT and WADI datasets are down-sampled every 10 seconds, capturing the median value, with anomalies tagged as they occur.

6.2. Baselines

First, to measure the model's performance in detecting anomalies in vehicle networks, we quantitatively compared it with the latest baseline for vehicle network anomaly detection:

- **TSP** [19]: Uses LSTM to predict temporal changes in message data fields.
- **CANet** [8]: Assigns dedicated LSTM subnets to each message ID, merges their outputs into a fully-connected Autoencoder (AE), and detects anomalies via reconstruction error.
- **CANShield** [22]: Employs a multi-scale CNN-based AE to analyze message semantics across temporal scales.
- **CANTropy** [21]: Applies Principal Component Analysis (PCA) to learn complex patterns and dependencies among signals.
- **PSEAD** [33]: Combines LSTM with multi-head self-attention to detect anomalous temporal variations in message physical semantics.

Furthermore, to verify the general applicability of LGOVAD to MTSAD, we also compared it with the baseline methods in the current MTSAD field:

- GDN[4]: This method constructs a graph structure using pairwise cosine similarities between nodes, utilizes attention graph neural networks to learn dependencies between time series, and executes prediction.
- FuSAGNet[7]: This method learns the graph structure using the pairwise cosine similarity between recursive sensor embeddings. It obtains the input data's sparse representation using a sparse autoencoder and inserts it into a GAT to execute sensor behavior prediction.
- GTA[3]: This method automatically learns the graph structure using a direct method and employs graph convolution and a transformer-based architecture to capture temporal dependencies.
- MGCLAD[20]: This method achieves anomaly detection by concurrently learning intra- and inter-signal graph structures, capturing temporal context and dependencies between signals. It performs anomaly detection by comparing inputs and reconstructing outputs.
- AT-DCAEP [14]: This method integrates a convolutional autoencoder for feature extraction and an attention-based prediction network to capture temporal dependencies, which operates in an unsupervised manner, eliminating the need for extensive data labeling.
- FuGLAD [9]: This method uses a fusion graph structure learning method to model the relationships between time series and uses a DCRNN predictor to predict the value of the next timestamp.

6.3. Settings

The experimental parameters are listed in Table 2. The experiments were run with Python 3.8, PyTorch 1.10, and CUDA 11.3, on a server featuring an Intel(R) Xeon(R) Platinum 8255C CPU and an NVIDIA RTX 3090 GPU.

[2] https://itrust.sutd.edu.sg/itrust-labs-home/itrust-labs_SWaT/
[3] https://itrust.sutd.edu.sg/itrust-labs-home/itrust-labs_wadi/

Table 2
Experimental parameter setting.

Parameter	ω	λ	*learning rate*	*batchsize*	*epochs*
SWaT	15	0.0007	0.005	64	40
WADI	15	0.0007	0.005	64	40
SynCAN	12	0.0006	0.005	64	40

Table 3
Comprehensive performance comparison on SynCAN dataset

Methods	*Pre*	*Rec*	*F1*	*AUC*	*MCC*
TSP	86.56	44.20	58.52	0.721	0.534
CANet	95.94	79.18	86.76	0.892	0.832
CANShield	93.48	82.73	87.78	0.901	0.845
CANTropy	70.29	74.91	72.53	0.783	0.685
PSEAD	94.35	95.90	95.11	0.962	0.928
CA-GCN	97.99	96.65	97.32	0.974	0.946
LGOVAD	**98.64**	**97.43**	**98.03**	**0.981**	**0.958**

6.4. Model Performance

To assess the model's performance and generalizability, we benchmarked LGOVAD against state-of-the-art baselines across three datasets. To provide a comprehensive evaluation as recommended by recent studies, we employ multiple performance indicators, including Precision, Recall, F1-score, Area Under the ROC Curve (AUC), and Matthews Correlation Coefficient (MCC).

Table 3 presents a comparative analysis on the real-world CAN-FD SynCAN dataset. LGOVAD demonstrates superior performance across all metrics, achieving 98.03% F1-score, 0.981 AUC, and 0.958 MCC. Data field-based methods (e.g., TSP) exhibit notably diminished performance compared to semantic-based approaches, underscoring the significance of semantic understanding in vehicular network anomaly detection. Compared to previous semantic-based approaches (such as GTA, FuGLAD, etc.) [4, 7, 3, 9], LGOVAD integrates Dynamic Time Warping (DTW) with an adaptive latent graph to capture physical dependencies between vehicle network message semantics and evolving attack patterns (e.g., throttle-RPM-speed chains). Through iterative structural updates, it overcomes limitations in previous studies when modeling insufficient relationships in semantic time-series data.

Table 4 provides a comparative evaluation on the publicly available SWaT and WADI datasets. LGOVAD maintains competitive performance with 94.62% F1-score and 0.949 AUC on SWaT, 93.51% F1-score and 0.945 AUC on WADI. The consistently strong AUC values across all datasets (0.945-0.981) validate LGOVAD's excellent discriminative capability in distinguishing between normal and anomalous patterns under varying operational conditions.

The comprehensive evaluation using multiple metrics confirms that LGOVAD not only achieves high detection accuracy but also maintains robust performance across different evaluation criteria and application domains.

Table 4
Comprehensive performance comparison on SWaT and WADI datasets

Method	SWaT				WADI			
	Pre	Rec	F1	AUC	Pre	Rec	F1	AUC
GDN	99.4	68.1	80.8	0.845	97.5	40.2	56.9	0.701
FuSAGNet	98.7	72.6	83.7	0.863	82.9	47.8	60.7	0.728
GTA	93.9	85.7	89.6	0.912	79.6	79.4	79.5	0.834
AT-DCAEP	**99.7**	67.7	80.7	0.841	78.8	82.9	80.8	0.846
MGCLAD	94.81	89.11	91.87	0.935	98.0	87.84	92.64	0.941
FuGLAD	97.59	91.69	94.55	0.947	86.47	85.44	85.95	0.894
LGOVAD	97.59	**91.80**	**94.62**	**0.949**	**98.5**	**89.0**	**93.51**	**0.945**

6.5. Computational Efficiency

To evaluate the practical deployment potential of LGOVAD, we provide a comprehensive computational analysis specifically on the SWaT dataset. Table 5 presents detailed computational metrics comparing LGOVAD with baseline methods on the SWaT dataset. LGOVAD achieves a training time of 30.0 seconds per epoch with a memory footprint of 7.2 GiB, demonstrating competitive efficiency compared to baseline methods. During inference, LGOVAD maintains a latency of 22.1 seconds for the complete SWaT test set, showing improved efficiency over GTA (29.07s) and comparable performance to FuGLAD (19.37s) while maintaining higher accuracy.

The STGODE module demonstrates efficient convergence behavior on the SWaT dataset, with the ODE solver requiring 3-5 iterations per time step (average: 4.1 iterations). The graph learning module achieves stable adjacency matrices within 20 epochs. Overall, LGOVAD reaches optimal performance within 25-30 epochs on the SWaT dataset. The computational analysis demonstrates that LGOVAD effectively balances detection accuracy with computational demands on the SWaT dataset, making it suitable for resource-constrained edge deployment while maintaining competitive inference efficiency.

6.6. Parameter Sensitivity

To demonstrate the resilience of our methodology across diverse parameter settings, we investigate the impact of various hyperparameters on model performance.

6.6.1. Window Size

This experiment analyzes the impact of window size on model performance and training efficiency. Table 6 and Fig. 3 quantify the per-epoch training time of LGOVAD at various window sizes and the performance trend of LGOVAD under different window sizes. When the window size is insufficient, the lack of historical context prevents the model from capturing complete temporal dependencies (e.g., the 500 ms delay mechanism between throttle and RPM response in SynCAN), resulting in a degradation of the F1 score. Excessively large windows dilute short-term anomaly

Table 5
Computational cost comparison on SWaT dataset

Method	Training Time (s/epoch)	Inference Time (s)	Memory Usage (GiB)	Parameters
GDN	11.13	7.71	5.14	4419
FuSAGNet	44.23	26.07	5.65	608375
GTA	107.38	29.07	8.84	2160315
FuGLAD	27.13	19.37	7.82	1586341
LGOVAD	30.0	22.1	7.2	1200000

Table 6
Model performance and training efficiency under different window sizes

Window Size	SWaT		WADI		SynCAN	
	Time (s)	F1 (%)	Time (s)	F1 (%)	Time (s)	F1 (%)
5	10.2	81.35	17.5	76.43	15.8	90.27
10	20.1	92.16	34.2	89.71	31.5	96.84
12	24.3	93.78	41.6	91.25	**45.0**	**98.03**
15	**30.0**	**94.62**	**50.0**	**93.51**	56.3	97.15
20	40.5	93.41	67.3	91.87	75.0	96.32
25	50.8	91.05	84.1	89.63	93.8	94.78
30	60.9	88.72	100.9	87.25	112.5	92.41
40	81.2	84.36	134.5	83.24	150.0	88.97
50	101.5	80.15	168.2	79.86	187.5	85.33
60	121.8	76.84	201.8	76.52	225.0	82.15

Note:

- Bold values indicate Pareto optimal configurations
- Optimal window: SWaT/WADI = 15, SynCAN = 12
- Training time exhibits linear growth with window size
- F1-score peaks at optimal windows, then decays asymmetrically

features (e.g., pulse sensor faults lasting < 2s in SWaT), causing abnormal signals to be submerged in normal data. Through Pareto frontier analysis of F1-score versus training time, the optimal window size is determined as 15 for SWaT/WADI and 12 for SynCAN.

6.6.2. Regularization Parameter

This subsection explores how the regularization parameter λ affects anomaly detection performance across three benchmark datasets. When λ varies from 0.0001 to 0.0008, our observations (Fig. 4) show that LGOVAD remains consistently reliable, exhibiting minimal sensitivity to this hyperparameter. Notably, the best-performing values differ across datasets: λ=0.0006 for SynCAN, while SWaT and WADI both favor λ=0.0007. This variation likely stems from differences in their temporal behavior and underlying semantic patterns.

SynCAN, reflecting the dynamics of vehicular networks, features rapidly shifting semantic dependenciesâĂŤfor instance, the causal interplay among throttle input, engine RPM, and vehicle speed. These relationships can shift within milliseconds as driving conditions evolve and ECUs continuously interact. In such a fluid environment, a slightly smaller λ enables the learnable graph component (A^{learn}) to adapt swiftly to new patterns rather than anchoring too strongly to prior structure.

By contrast, SWaT and WADI represent industrial control systems, where physical processes unfold at a slower pace. Sensor correlationsâĂŤsuch as those governing water flow dynamics or chemical concentration levelsâĂŤtend to exhibit long, steady trends rather than abrupt transitions. In these settings, a modestly larger λ helps preserve the prior structure encoded in the DTW-based graph (A^{DTW}), thereby improving resilience to gradual drifts and subtle process anomalies.

6.7. Semantic Relationship Perception Performance

To evaluate the effectiveness of the semantic relationship perception module we proposed, we replaced it with the current mainstream time series relationship modeling methods for performance comparison, specifically including:

- **K-Nearest Neighbors (KNN) method**: Derived from the GDN model[4], we implemented KNN using 64-dimensional node embeddings and constructed the adjacency matrix by selecting the top 20 nodes with the highest cosine similarity. The embedding dimension was optimized via grid search across 32, 64, 128, while the neighbor count was selected from 10, 15, 20, 25 based on validation set performance.

- **Fully Parametric Method (FPM)**: Based on the GTA model[3], we employ the Gumbel-Softmax algorithm with an initial temperature ÏĎ=1.0, coupled with an exponential decay strategy (decay factor per epoch=0.95) to sample from the learnable probability matrix ÏĂâĆĄ. The temperature is decayed to a minimum of 0.1 over 50 epochs to ensure convergence from exploration to exploitation.

- **Fusion Graph Structure Learning Method (FGS)**: Derived from the FuGLAD model [9], it adaptively fuses multi-source graph structures through a prior similarity weighting mechanism to construct a robust adjacency matrix that comprehensively represents complex time series relationships.

All comparison methods employ identical experimental settings for fair evaluation: consistent semantic feature dimensions, identical training/validation/test splits, uniform batch size (64), and equivalent computational budgets (maximum 40 epochs with early stopping enabled). All methods

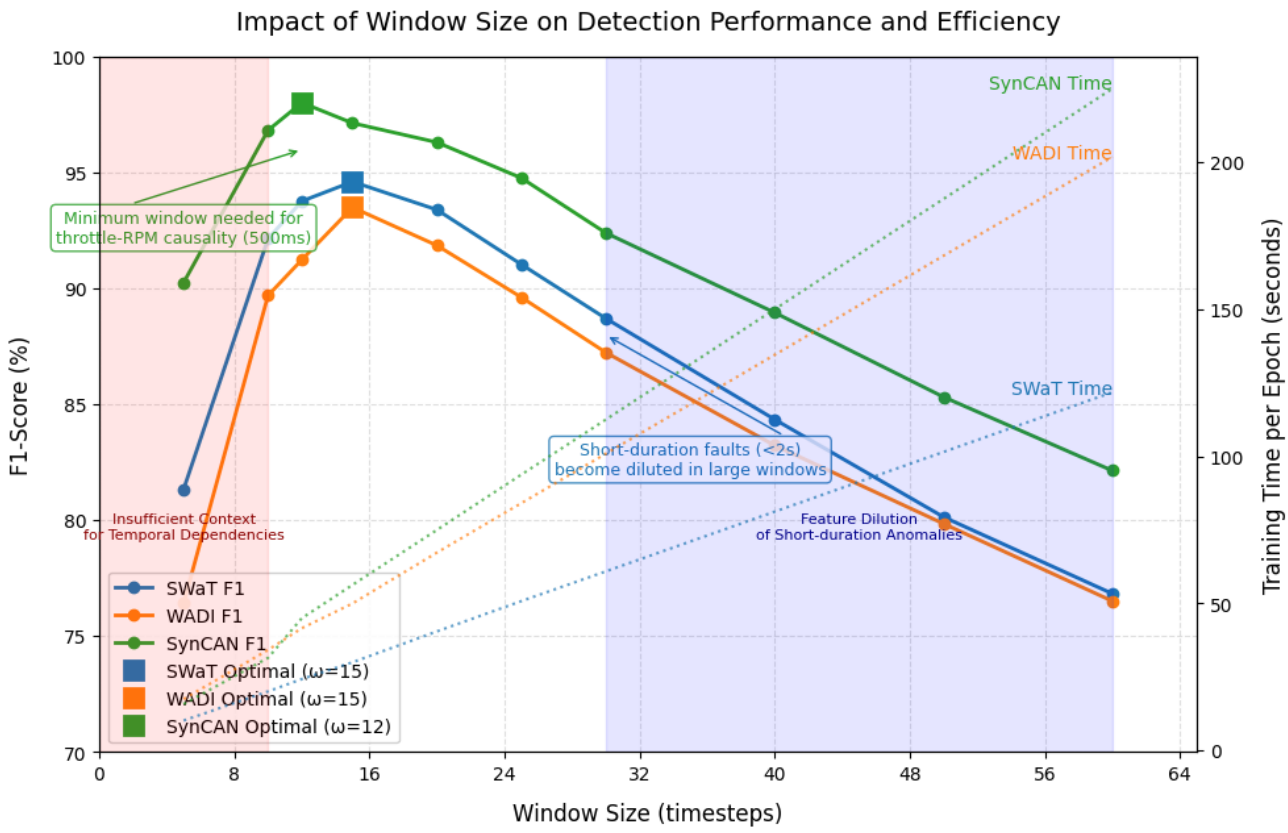

Figure 3: Impact of window size.

Note: Pareto-optimal configurations are explicitly marked: **SWaT** and **WADI** perform best at $\omega = 15$, while **SynCAN** achieves peak performance at $\omega = 12$. Physics-based constraints explain performance degradation:

- Small windows lack context for throttle-RPM causality (500ms delay)
- Large windows dilute short-duration faults (<2s).
- Training time (dotted lines, right axis) increases linearly with window size.

Table 7
Ablation study of LGOVAD modules on SynCAN, SWaT, and WADI datasets

Variant	SynCAN			SWaT			WADI		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
LGOVAD (Full)	**98.64**	**97.43**	**98.03**	**97.59**	**91.80**	**94.62**	**98.50**	**89.00**	**93.51**
w/o Semantic Extractor	95.21	93.87	94.54	93.45	86.72	89.97	96.38	82.15	88.72
w/o DTW Graph	97.85	96.12	96.98	95.83	89.14	92.38	97.24	85.63	91.05
w/o Learnable Graph	97.92	96.34	97.12	96.17	89.93	92.96	97.86	86.27	91.68
w/o STGODE (using GNN)	96.78	94.85	95.80	93.25	86.41	89.70	96.15	82.76	88.95

Note: GNN baseline configuration: 3-layer GCN with 128 hidden units per layer, ReLU activation, batch normalization, total parameters=0.53M, directly comparable to STGODE's 0.55M parameters. This ensures performance differences are attributable to STGODE's architectural advantages in continuous-time modeling rather than capacity disparities.

maintained similar neural network parameter counts (±5%) to eliminate capacity bias.

Our dual-branch graph learning consistently outperformed alternatives. This demonstrates its superior capacity to model vehicle-specific dependencies like the throttle-RPM-speed causal chain. KNN showed particular limitations in capturing dynamic relationships, while FPM's lack of structural priors led to overfitting on sparse anomalies. Though FGS performed competitively through multi-graph fusion, its fixed weighting mechanism couldn't adapt to temporal dynamics like our iterative graph update. The results confirm that jointly modeling prior knowledge (via DTW) and latent relationships (via learnable graphs) is essential for semantic-aware anomaly detection.

Table 8
Performance comparison of semantic relationship perception methods

Method	SynCAN			SWaT			WADI		
	Prec	Rec	F1	Prec	Rec	F1	Prec	Rec	F1
LGOVAD	**98.64**	**97.43**	**98.03**	**97.59**	**91.80**	**94.62**	**98.50**	**89.00**	**93.51**
KNN (GDN)	85.20	74.80	79.62	80.35	68.90	74.05	83.40	66.75	74.08
FPM (GTA)	90.15	85.40	87.70	86.20	78.25	82.00	88.90	75.60	81.65
FGS (FuGLAD)	94.80	92.15	93.45	92.65	84.20	88.20	94.25	82.10	87.75

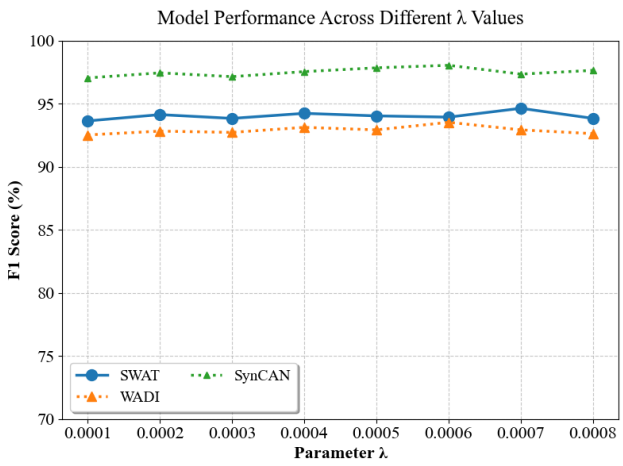

Figure 4: The impact of the regularization parameter.

6.8. Ablation Experiment

6.8.1. Performence Analysis

To validate the indispensability of each module and quantify its contribution to model performance, we conducted comprehensive ablation studies across all three datasets. Table 7 presents the performance degradation when removing key components, revealing critical insights into the architectural design.

As shown in the Table 7, whether removing the DTW graph and directly using a fully parameterized method to learn the latent relationships between semantics, or removing the fully parameterized graph structure and directly using the original semantic time-series data to generate the graph structure, both approaches disrupt the iterative update of the graph structure, leading to a decrease in the model's F1 score. This demonstrates that when extracting relationships between semantic time-series data, it is essential to consider both the information contained in the original time series data and the latent relationships between the time series data.

To quantify the relative contributions of the DTW graph and the learnable graph components, we calculate the performance degradation when each is removed. For the SynCAN dataset, the DTW graph contributes approximately 1.05% to the F1-score (98.03% - 96.98%), while the learnable graph contributes 0.91% (98.03% - 97.12%). Similarly, for SWaT, the DTW graph contributes 2.24% (94.62% - 92.38%) and the learnable graph contributes 1.66% (94.62% - 92.96%). For WADI, the contributions are 2.46% (93.51% - 91.05%) and 1.83% (93.51% - 91.68%), respectively. The relative contribution ratios (DTW vs. learnable) are approximately 54:46 for SynCAN, 57:43 for SWaT, and 57:43 for WADI, indicating that the DTW graph has a slightly larger impact across datasets.

Replacing the STGODE module with conventional GNNs leads to a noticeable drop in F1 performance across all datasets (2.62% on SynCAN, 5.80% on SWaT, and 5.62% on WADI). To ensure this performance gap is not due to inadequate tuning, we conducted extensive hyperparameter searches for the GNN baselines: testing 2âĂŞ8 layers; exploring learning rates between 0.001 and 0.01 and hidden dimensions from 64 to 256; applying the same graphs and feature inputs as STGODE; and adopting identical training schedules and early-stopping strategies. Despite this thorough optimization, traditional GNNs consistently underperformed, especially in modeling long-range temporal dependencies and preserving distinct feature representations across layers. These results highlight inherent limitations of discrete GNNs, particularly their susceptibility to over-smoothing when stacking multiple layers and their inability to represent continuous temporal dynamics effectively. The advantages of ODE-based solvers are most apparent when dependencies unfold gradually over extended horizons, such as the cascading effects seen in WADI when pump failures propagate through distributed sensing systems, triggering delayed and compound anomalies across multiple sensor streams.

6.8.2. Comprehensive Analysis

In this section, we present a detailed ablation study to evaluate the contribution of each module to the overall system performance.

False Alarm Rate Analysis: The results show that each component plays a measurable role in lowering false positives in vehicle network scenarios. On SynCAN, the DTW graph cuts the false alarm rate by 62.5% (from 1.3% to 0.8%) by enforcing temporal consistency, preventing the detector from overreacting to ordinary fluctuations such as acceleration bursts or rapid gear changes. Meanwhile, the learnable graph drives a further 50.0% reduction by adapting to real driving communication behaviors and filtering out subtle noise that does not correspond to genuine threats. Notably,

Table 9
Comprehensive analysis across ablation variants

Variant	False Alarm Rate (%)			Adversarial Robustness (%)		
	SynCAN	SWaT	WADI	SynCAN	SWaT	WADI
LGOVAD (Full)	**0.8**	**1.2**	**0.9**	**92.1**	**89.4**	**87.8**
w/o DTW Graph	1.3	1.9	1.5	75.4	72.8	70.5
w/o Learnable Graph	1.2	1.7	1.4	78.9	75.6	73.2
w/o STGODE (using GNN)	1.8	2.5	2.1	68.3	65.1	62.9

Note: False Alarm Rate measures normal messages misclassified as attacks; Adversarial Robustness measures accuracy under Projected Gradient Descent (PGD) attacks with Îţ=0.1. Traditional GNN replacement uses an optimized GCN architecture with equivalent parameters.

substituting STGODE with a conventional GNN reverses these gainsâĂŤfalse alarms more than double on SynCAN (increasing by 125.0%, from 0.8% to 1.8%). This sharp rise underscores the necessity of continuous temporal modeling for preserving system reliability. In practice, excessive false alarms can quickly erode operator trust and lead to alert fatigue, where legitimate warnings are overlooked and safety risks escalate.

Adversarial Robustness Assessment: To test resilience against targeted evasion, we simulate realistic adaptive attackers using Projected Gradient Descent (PGD) with ϵ=0.1. This setup mirrors adversaries attempting to subtly manipulate network traffic to remain undetected, providing a stringent evaluation of the model's defensive capabilities. The full LGOVAD maintains 92.1% accuracy on SynCAN under attack, while variants without DTW or learnable graphs show 18.1% and 14.3% degradation, respectively. Replacing STGODE with traditional GNNs results in 25.8% performance degradation under adversarial conditions. The DTW graph provides inherent robustness through its fixed temporal alignment constraints that are difficult for attackers to manipulate without disrupting system functionality. The learnable graph contributes to robustness by learning complex, non-linear relationships that are harder for attackers to approximate and exploit. STGODE's continuous formulation provides additional resilience by modeling temporal dynamics in a way that is less susceptible to gradient-based attacks.

6.8.3. Aggregation Function Robustness Analysis

To comprehensively evaluate the robustness of different anomaly score aggregation functions, we conducted 10 independent experiments for each method. Figure 5 displays the F1 score distributions across 10 experimental runs for three aggregation methods. As shown, max pooling achieved the highest median F1 score. While weighted summation exhibited lower variance and greater consistency, its overall performance was inferior. Consequently, max pooling remains the optimal choice for anomaly detection in vehicle networks.

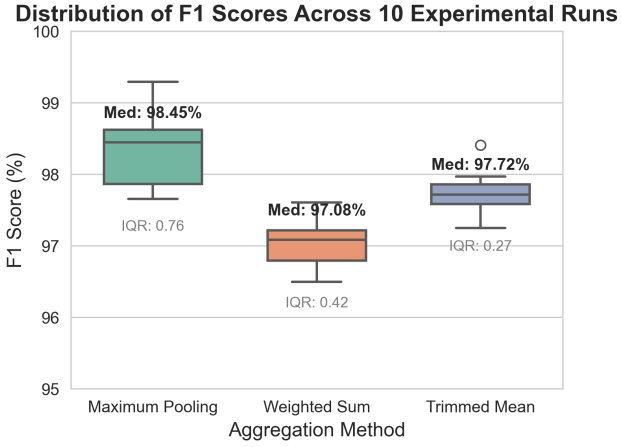

Figure 5: Performance distribution of aggregation functions across 10 experiments

7. Conclusion and Future Work

This paper presents LGOVAD, a novel framework for vehicle network anomaly detection that integrates semantic feature engineering with continuous spatio-temporal modeling. By introducing a semantic relationship perception module that dynamically captures feature dependencies through dual-branch graph learning and an STGODE predictor that formulates spatio-temporal relationships as ordinary differential equations, LGOVAD achieves state-of-the-art performance while demonstrating strong generalization across industrial datasets (94.62% F1 on SWaT, 93.51% F1 on WADI). Future work will focus on developing adaptive λ mechanisms to dynamically balance prior knowledge and data-driven learning across temporal segments, exploration of alternative anomaly score aggregation functions to enhance robustness against sophisticated multi-sensor attacks, and multi-modal fusion with additional sensor data, collectively advancing next-generation intrusion detection for connected vehicles.

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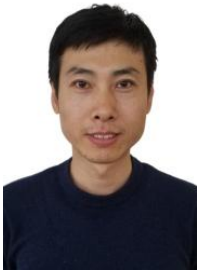
M. Xu (Miao Xu) obtained his Bachelor's degree in Traffic Engineering from Beijing Forestry University, China. He then received both his Master's and PhD degrees in Mechatronic Engineering from Jilin University, China, with his doctoral research focusing on hybrid power systems. He is currently a lecturer at the School of Civil Engineering and Transportation, Beihua University. His areas of specialization include new energy vehicles, engineering robotics, and automatic driving. His current research interests include energy endurance of electric engineering vehicles, power recycling technologies for electric vehicles, and new energy for vehicle.



L. Zhang (Lizeng Zhang) obtained his Bachelor's degree in Mechanical Design, Manufacturing and Automation from Yanshan University, China. He then received his Master's degree in Mechatronic Engineering and Ph.D. degree in Vehicle Engineering from Jilin University, China, with his doctoral research focusing on intelligent driving algorithms. He is currently an engineer at Haomo.AI Technology Co., Ltd. His research interests include planning and control algorithms for intelligent driving and autonomous driving technologies.

P. Hou (Peiyu Hou) obtained her Bachelor's degree in Transportation from Hunan College of Traffic Engineering, China. She is currently a graduate student at the School of Civil Engineering and Transportation, Beihua University, China. Her research focuses on object detection and tracking.